Resolution Rule

in Predicate Logic

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Outline

1. Substitution
2. Unification
3. Most General Unifier (**mgu**)
4. Resolution rule
5. Applications

Substitution

A substitution of terms for variables is a set:

$x_1 \leftarrow t_1, ..., x_n \leftarrow t_n$

Where,

- Each x_i is a distinct variable.
- Each t_i is a term which is not identical to the corresponding variable x_i.
- The empty substitution is the empty set.

Substitution: Notation

- Substitutions are denoted by lLower-case Greek letters: $\lambda, \mu, \sigma, \theta$
- The empty substitution is denoted by ϵ.

Substitution

Expression

An expression is a term, a literal, a clause or a set of clauses.

Substitution on expression

- Let E be an expression
- Let $\theta = x_1 \leftarrow t_1, ..., x_n \leftarrow t_n$ be a substitution.
- $E\theta$ means an application of θ on E.

$E\theta$: is obtained by simultaneously replacing each occurrence of x_i in E by t_i.

Substitution: Example

$$
\begin{aligned}
E &= p(x), q(f(y)) \\
\theta &= x \leftarrow y, y \leftarrow f(a) \\
& \text{then} \\
E\theta &= p(y), q(f(f(a)))
\end{aligned}
$$

Substitution

Composition of Substitution

- $\theta = \{x_1 \leftarrow t_1, ..., x_n \leftarrow t_n\}$
- $\sigma = \{y_1 \leftarrow s_1, ..., y_k \leftarrow s_k\}$
- $X = \{x_1, ..., x_n\}$: set of variables in θ
- $Y = \{y_1, .., y_k\}$: set of variables in σ
- $\theta\sigma$: Composition of θ and σ

$$\theta\sigma = \{x_i \leftarrow t_i\sigma | x_i \in X, x_i \neq t_i\sigma\} \cup \{y_j \leftarrow s_j | y_j \in Y, y_j \notin X\}$$

Meaning: apply the substitution σ to the terms t_i of θ (provided that the resulting substitutions do not collapse to $x_i \leftarrow x_i$) and then append the substitutions from σ whose variables do not already appear in θ.

Composition of Substitution: Example

$$
\begin{aligned}
E &= p(u,v,x,y,z)\\
\theta &= \{x \leftarrow f(y), y \leftarrow f(a), z \leftarrow u\},\\
\sigma &= \{y \leftarrow g(a), u \leftarrow z, v \leftarrow f(f(a))\},\\
&\ \text{then}\\
\theta\sigma &= \{x \leftarrow f(g(a)), y \leftarrow f(a), u \leftarrow z, v \leftarrow f(f(a))\}\\
E(\theta\sigma) &= \{p(z, f(f(a)), f(g(a)), f(a), z)\}\\
E\theta &= \{p(u, v, f(y), f(a), u)\}\\
(E\theta)\sigma &= \{p(z, f(f(a)), f(g(a)), f(a), z)\}\\
&= E(\theta\sigma)
\end{aligned}
$$

Composition of Substitution: Lemmas

Lemma

For any expression E and substitutions θ , σ , $E(\theta\sigma) = (E\theta)\sigma$.

Lemma

For any substitutions θ , σ , λ, $\theta(\sigma\lambda) = (\theta\sigma)\lambda$.

Unification

- Let $E_1 = \{p(f(x), g(y))\}$
- Let $E_2 = \{p(f(f(a)), g(z))\}$
- Let $\theta = \{x \leftarrow f(a), y \leftarrow f(g(a)), z \leftarrow f(g(a))\}$

E_1 and E_2 are different.
However, they can be unified by the substitution θ. That is $E_1\theta = E_2\theta$.
$E_1\theta = E_2\theta = \{p(f(f(a)), g(f(g(a))))\}$

Unification

Unifier

Let $U = \{A_1, ..., A_n\}$ be a set of atoms. A unifier θ is a substitution such that: $A_1\theta = = A_n\theta$

Most general unifier (mgu)

A most general unifier (mgu) for U is a unifier μ such that any unifier θ of U canbe expressed as: $\theta = \mu\lambda$
for some substitution λ.

$$
\begin{aligned}
A &= \{p(f(x), g(y)), p(f(f(a)), g(z))\} \\
\theta_1 &= \{x \leftarrow f(a), y \leftarrow f(g(a)), z \leftarrow f(g(a))\}, \\
\theta_2 &= \{x \leftarrow f(a), y \leftarrow a, z \leftarrow a\}, \\
\mu &= \{x \leftarrow f(a), z \leftarrow y\}, \\
& \text{then} \\
& \mu \text{ is a mgu} \\
\theta_1 &= \mu\{y \leftarrow f(g(a))\}, \\
\theta_2 &= \mu\{y \leftarrow a\}, \\
\\
A\theta_1 &= \ ? \\
A\theta_2 &= \ ? \\
A\mu &= \ ?
\end{aligned}
$$

Unification

Disagreement Set

- Let A_1 and A_2 be two atoms with the same predicate symbols.Considering them as sequences of symbols.
- Let k be the leftmost position at which the sequences are different.
- Let t_1 be the term beginning at k in sequence A_1.
- Let t_2 be the term beginning at k in sequence A_2.
- Then:
  $D = \{t_1, t_2\}$ is called Disagreement Set of A_1 and A_2.

$$
\begin{array}{rl}
A_1 = & p(g(y), f(x, h(x), y)) \\
A_2 = & p(x, f(g(z), w, z)) \\
 & \text{then} \\
D = & \{x, g(y)\}
\end{array}
$$

Robinson's Unification Algorithm

- **INPUT**:
  - A_1 and A_2 be two atoms with the same predicate symbols.
- **OUTPUT**: one of the two cases below
  - An mgu for A_1 and A_2 if they can be unified.
  - Report that they are not unifiable.

1. Let $i = 0$; Assign $\sigma_i = \varnothing$; Assign $A_1^i = A_1$ and $A_2^i = A_2$;
2. Repeat the following steps constantly
   1. Compute disagreement set $D_i = \{t_1, t_2\}$ for A_1^i and A_2^i.
   2. If D_i is empty then return σ_i
   3. Otherwise: If one term is a variable x_{i+1} and the other is a term t_{i+1} such that x_{i+1} does not occur in t_{i+1} then
      - $\sigma_{i+1} = \{x_{i+1} \leftarrow t_{i+1}\}$; $A_1^{i+1} = A_1^i \sigma_{i+1}$; $A_2^{i+1} = A_2^i \sigma_{i+1}$
      - $i = i + 1$
   4. Otherwise: Otherwise: Report not unifiable.

Step.	Materials	Notes
0.	$\sigma_0 = \varnothing$	
	$A_1^0 = p(g(y), f(x, h(x), y))$	
	$A_2^0 = p(x, f(g(z), w, z))$	
1.	$D_0 = \{x, g(y)\}$	
	$\sigma_1 = \{x \leftarrow g(y)\}$	
	$A_1^1 = A_1^0\sigma_1 = p(g(y), f(g(y), h(g(y)), y))$	
	$A_2^1 = A_2^0\sigma_1 = p(g(y), f(g(z), w, z))$	
2.	$D_1 = \{y, z\}$	
	$\sigma_2 = \{y \leftarrow z\}$	
	$A_1^2 = A_1^2\sigma_2 = p(g(z), f(g(z), h(g(z)), z))$	
	$A_2^2 = A_2^2\sigma_2 = p(g(z), f(g(z), w, z))$	
3.	$D_2 = \{w, h(g(z))\}$	
	$\sigma_3 = \{w \leftarrow h(g(z))\}$	
	$A_1^3 = A_1^2\sigma_3 = p(g(z), f(g(z), h(g(z)), z))$	
	$A_2^3 = A_2^2\sigma_3 = p(g(z), f(g(z), h(g(z)), z))$	
4.	$D_3 = \varnothing \rightarrow$ STOP	
	$mgu = \mu = \sigma_0\sigma_1\sigma_2\sigma_3 = \{x \leftarrow g(z), y \leftarrow z, w \leftarrow h(g(z))\}$	

Resolution Rule

$$\frac{C_1, C_2}{\{C_1\sigma - L_1\sigma\} \cup \{C_2\sigma - L_2\sigma\}}$$

- C_1 and C_2 are clauses having no common variable.
- $L_1 = \{l_1^1, ..., l_1^{n_1}\} \subseteq C_1$
- $L_2 = \{l_2^1, ..., l_2^{n_2}\} \subseteq C_2$
- L_1 and L_2^c can be unified by mgu σ
- C_1 and C_2 are said to be clashing clauses on L_1 and L_2

Resolution Rule: Example

Action	Clause 1	Clause 2
Input	$C_1 = \{p(f(x), g(y)), q(x, y)\}$	$C_2 = \{\neg p(f(f(a)), g(z)), q(f(a), z)\}$
Select	$L_1 = \{p(f(x), g(y))\}$	$L_2 = \{\neg p(f(f(a)), g(z))\}$
		$L_2^c = \{p(f(f(a)), g(z))\}$
	L_1 and L_2^c are unifiable	
Compute mgu	$\sigma = \{x \leftarrow f(a), y \leftarrow z\}$	
	$L_1\sigma = L_2^c\sigma = \{p(f(f(a)), g(z))\}$	
Compute	$C_1\sigma = \{p(f(f(a)), g(z)), q(f(a), z)\}$	$C_2\sigma = \{\neg p(f(f(a)), g(z)), q(f(a), z)\}$
Compute	$C_1\sigma - L_1\sigma = \{q(f(a), z)\}$	$C_2\sigma - L_2\sigma = \{q(f(a), z)\}$
Compute resolvant	$\{C_1\sigma - L_1\sigma\} \cup \{C_2\sigma - L_2\sigma\} = \{q(f(a), z)\}$	

Resolution Rule: Example

Action	Clause 1	Clause 2
Input	$C_1 = \{p(f(x))\}$	$C_2 = \{\neg p(x)\}$
Rename variables	$C_1 = \{p(f(x))\}$	$C_2 = \{\neg p(y)\}$
Select	$L_1 = \{p(f(x))\}$	$L_2 = \{\neg p(y)\}$
		$L_2^c = \{p(y)\}$
	L_1 and L_2^c are unifiable	
Compute mgu	$\sigma = \{y \leftarrow f(x)\}$	
	$L_1\sigma = L_2^c\sigma = \{p(f(x))\}$	
Compute	$C_1\sigma = \{p(f(x))\}$	$C_2\sigma = \{\neg p(f(x))\}$
Compute	$C_1\sigma - L_1\sigma = \varnothing$	$C_2\sigma - L_2\sigma = \varnothing$
Compute resolvant	$\{C_1\sigma - L_1\sigma\} \cup \{C_2\sigma - L_2\sigma\} = \varnothing = \square$	

Resolution Rule: Example

Action	Clause 1	Clause 2
Input	$C_1 = \{p(x), p(y)\}$	$C_2 = \{\neg p(x), \neg p(y)\}$
Rename variables	$C_1 = \{p(x), p(y)\}$	$C_2 = \{\neg p(z), \neg p(t)\}$
Select	$L_1 = \{p(x), p(y)\}$	$L_2 = \{\neg p(z), \neg p(t)\}$
		$L_2^c = \{p(z), p(t)\}$
	L_1 and L_2^c are unifiable	
Compute mgu	$\sigma = \{y \leftarrow x, z \leftarrow x, t \leftarrow x\}$	
	$L_1\sigma = L_2^c\sigma = \{p(x)\}$	
Compute	$C_1\sigma = \{p(x)\}$	$C_2\sigma = \{\neg p(x)\}$
Compute	$C_1\sigma - L_1\sigma = \varnothing$	$C_2\sigma - L_2\sigma = \varnothing$
Compute resolvant	$\{C_1\sigma - L_1\sigma\} \cup \{C_2\sigma - L_2\sigma\} = \varnothing = \square$	